

Multivariate Analysis – Honours

Section 1 Exercises Solutions

Chapter 3

Homework exercise 3.1

Given: $\mu_1 = 0$, $\mu_2 = 2$, $\sigma_{11} = 2$, $\sigma_{22} = 1$, $\rho_{12} = 0.5$.

1.

$$\begin{aligned}
 f(x_1, x_2) &= \frac{1}{2\pi\sqrt{\sigma_{11}\sigma_{22}(1-\rho_{12}^2)}} \times e^{-\frac{1}{2(1-\rho_{12}^2)} \times \left[\left(\frac{x_1-\mu_1}{\sqrt{\sigma_{11}}} \right)^2 + \left(\frac{x_2-\mu_2}{\sqrt{\sigma_{22}}} \right)^2 - 2\rho_{12} \left(\frac{x_1-\mu_1}{\sqrt{\sigma_{11}}} \right) \left(\frac{x_2-\mu_2}{\sqrt{\sigma_{22}}} \right) \right]} \\
 &= \frac{1}{2\pi\sqrt{(2)(1)(1-\frac{1}{4})}} \times e^{-\frac{1}{2(1-\frac{1}{4})} \times \left[\left(\frac{x_1-0}{\sqrt{2}} \right)^2 + \left(\frac{x_2-2}{\sqrt{1}} \right)^2 - 2 \times \frac{1}{2} \times \left(\frac{x_1-0}{\sqrt{2}} \right) \left(\frac{x_2-2}{\sqrt{1}} \right) \right]} \\
 &= \frac{1}{2\pi\sqrt{\frac{3}{2}}} \exp \left\{ -\frac{1}{2} \times \left[\frac{2}{3}x_1^2 + \frac{4}{3}(x_2-2)^2 - \frac{2\sqrt{2}}{3}x_1(x_2-2) \right] \right\}
 \end{aligned}$$

2. From the above we can see that

$$(\mathbf{x} - \boldsymbol{\mu})' \boldsymbol{\Sigma}^{-1} (\mathbf{x} - \boldsymbol{\mu}) = \frac{2}{3}x_1^2 + \frac{4}{3}(x_2-2)^2 - \frac{2\sqrt{2}}{3}x_1(x_2-2)$$

Homework exercise 3.2

1. To prove: $(\mathbf{X} - \boldsymbol{\mu})' \boldsymbol{\Sigma}^{-1} (\mathbf{X} - \boldsymbol{\mu}) \sim \chi_p^2$

Method 1, using the spectral decomposition of $\boldsymbol{\Sigma}$:

We start by noting that since $\boldsymbol{\Sigma} \mathbf{e}_i = \lambda_i \mathbf{e}_i$ and $\boldsymbol{\Sigma}$ is positive definite, $\boldsymbol{\Sigma}^{-1} \mathbf{e}_i = \frac{1}{\lambda_i} \mathbf{e}_i$

$$\therefore \boldsymbol{\Sigma}^{-1} = \sum_{i=1}^p \frac{1}{\lambda_i} \mathbf{e}_i \mathbf{e}_i'$$

Now

$$\begin{aligned}
(\mathbf{X} - \boldsymbol{\mu})' \boldsymbol{\Sigma}^{-1} (\mathbf{X} - \boldsymbol{\mu}) &= (\mathbf{X} - \boldsymbol{\mu})' \sum_{i=1}^p \frac{1}{\lambda_i} \mathbf{e}_i \mathbf{e}_i' (\mathbf{X} - \boldsymbol{\mu}) \\
&= \sum_{i=1}^p \frac{1}{\lambda_i} (\mathbf{X} - \boldsymbol{\mu})' \mathbf{e}_i \mathbf{e}_i' (\mathbf{X} - \boldsymbol{\mu}) \\
&= \sum_{i=1}^p \frac{1}{\lambda_i} [\mathbf{e}_i' (\mathbf{X} - \boldsymbol{\mu})]^2 \\
&= \sum_{i=1}^p \left[\frac{1}{\sqrt{\lambda_i}} \mathbf{e}_i' (\mathbf{X} - \boldsymbol{\mu}) \right]^2 \\
&= \sum_{i=1}^p Z_i^2
\end{aligned}$$

Where the Z_i can be arranged as

$$\mathbf{Z} = \mathbf{A}(\mathbf{X} - \boldsymbol{\mu})$$

$$\text{if } \mathbf{A} = \begin{bmatrix} \frac{1}{\sqrt{\lambda_1}} \mathbf{e}_1' \\ \frac{1}{\sqrt{\lambda_2}} \mathbf{e}_2' \\ \vdots \\ \frac{1}{\sqrt{\lambda_p}} \mathbf{e}_p' \end{bmatrix}.$$

Noting that $\mathbf{X} - \boldsymbol{\mu} \sim N_p(\mathbf{0}, \boldsymbol{\Sigma})$, we have

$$\mathbf{Z} = \mathbf{A}(\mathbf{X} - \boldsymbol{\mu}) \sim N_p(\mathbf{0}, \mathbf{A}\boldsymbol{\Sigma}\mathbf{A}')$$

Consider $\mathbf{A}\boldsymbol{\Sigma}$:

$$\mathbf{A}\boldsymbol{\Sigma} = \begin{bmatrix} \frac{1}{\sqrt{\lambda_1}} \mathbf{e}_1' \\ \frac{1}{\sqrt{\lambda_2}} \mathbf{e}_2' \\ \vdots \\ \frac{1}{\sqrt{\lambda_p}} \mathbf{e}_p' \end{bmatrix} \sum_{i=1}^p \lambda_i \mathbf{e}_i \mathbf{e}_i'$$

For each row $j = 1, \dots, p$ of $\mathbf{A}$, the product yields

$$\frac{1}{\sqrt{\lambda_j}} \mathbf{e}_j' \lambda_j \mathbf{e}_j \mathbf{e}_j' + \sum_{i \neq j} \frac{1}{\sqrt{\lambda_j}} \mathbf{e}_j' \lambda_i \mathbf{e}_i \mathbf{e}_i' = \sqrt{\lambda_j} \mathbf{e}_j'$$

since the $\mathbf{e}_i$ are mutually orthonormal.

Therefore,

$$\begin{aligned}
\mathbf{A}\mathbf{\Sigma}\mathbf{A}' &= \begin{bmatrix} \frac{1}{\sqrt{\lambda_1}}\mathbf{e}'_1 \\ \frac{1}{\sqrt{\lambda_2}}\mathbf{e}'_2 \\ \vdots \\ \frac{1}{\sqrt{\lambda_p}}\mathbf{e}'_p \end{bmatrix} \sum_{i=1}^p \lambda_i \mathbf{e}_i \mathbf{e}'_i \begin{bmatrix} \frac{1}{\sqrt{\lambda_1}}\mathbf{e}_1 & \frac{1}{\sqrt{\lambda_2}}\mathbf{e}_2 & \cdots & \frac{1}{\sqrt{\lambda_p}}\mathbf{e}_p \end{bmatrix} \\
&= \begin{bmatrix} \sqrt{\lambda_1}\mathbf{e}'_1 \\ \sqrt{\lambda_2}\mathbf{e}'_2 \\ \vdots \\ \sqrt{\lambda_p}\mathbf{e}'_p \end{bmatrix} \begin{bmatrix} \frac{1}{\sqrt{\lambda_1}}\mathbf{e}_1 & \frac{1}{\sqrt{\lambda_2}}\mathbf{e}_2 & \cdots & \frac{1}{\sqrt{\lambda_p}}\mathbf{e}_p \end{bmatrix} \\
&= \mathbf{I}
\end{aligned}$$

Now, since $\mathbf{Z} \sim N_p(\mathbf{0}, \mathbf{I})$, we have i.i.d. $Z_i \sim N(0, 1)$, $i = 1, \dots, p$.

$$\therefore (\mathbf{X} - \boldsymbol{\mu})' \boldsymbol{\Sigma}^{-1} (\mathbf{X} - \boldsymbol{\mu}) = \sum_{i=1}^p Z_i^2 \sim \chi_p^2$$

from which it naturally follows that

$$\Pr [(\mathbf{X} - \boldsymbol{\mu})' \boldsymbol{\Sigma}^{-1} (\mathbf{X} - \boldsymbol{\mu}) \leq \chi_p^2(\alpha)] = 1 - \alpha$$

Method 2, using the eigendecomposition of $\boldsymbol{\Sigma}$:

Since $\boldsymbol{\Sigma}$ is symmetric positive definite, we have

$$\begin{aligned}
\boldsymbol{\Sigma} &= \mathbf{U} \boldsymbol{\Lambda} \mathbf{U}' \\
\boldsymbol{\Lambda} &= \mathbf{U}' \boldsymbol{\Sigma} \mathbf{U},
\end{aligned}$$

where $\mathbf{U} = [\mathbf{e}_1 \quad \mathbf{e}_2 \quad \cdots \quad \mathbf{e}_p]$ is an orthogonal matrix such that $\mathbf{U}' = \mathbf{U}^{-1}$, and $\boldsymbol{\Lambda} = \text{diag}(\lambda_1, \dots, \lambda_p)$.

Consider

$$\begin{aligned}
(\mathbf{X} - \boldsymbol{\mu})' \boldsymbol{\Sigma}^{-1} (\mathbf{X} - \boldsymbol{\mu}) &= (\mathbf{X} - \boldsymbol{\mu})' (\mathbf{U} \boldsymbol{\Lambda} \mathbf{U}')^{-1} (\mathbf{X} - \boldsymbol{\mu}) \\
&= (\mathbf{X} - \boldsymbol{\mu})' \mathbf{U} \boldsymbol{\Lambda}^{-1} \mathbf{U}' (\mathbf{X} - \boldsymbol{\mu})
\end{aligned}$$

If we let $\mathbf{Y} = \boldsymbol{\Lambda}^{-\frac{1}{2}} \mathbf{U}' (\mathbf{X} - \boldsymbol{\mu})$, then

$$\boldsymbol{\Sigma}_Y = \boldsymbol{\Lambda}^{-\frac{1}{2}} \mathbf{U}' \boldsymbol{\Sigma} \mathbf{U} \boldsymbol{\Lambda}^{-\frac{1}{2}} = \boldsymbol{\Lambda}^{-\frac{1}{2}} \boldsymbol{\Lambda} \boldsymbol{\Lambda}^{-\frac{1}{2}} = \mathbf{I},$$

such that $\mathbf{Y} \sim N_p(\mathbf{0}, \mathbf{I})$.

I.e. $Y_i \stackrel{\text{iid}}{\sim} N(0, 1)$, $i = 1, \dots, p$.

Noting that $\boldsymbol{\Lambda}^{\frac{1}{2}} \mathbf{Y} = \mathbf{U}' (\mathbf{X} - \boldsymbol{\mu})$, we have

$$\begin{aligned}
(\mathbf{X} - \boldsymbol{\mu})' \boldsymbol{\Sigma}^{-1} (\mathbf{X} - \boldsymbol{\mu}) &= (\mathbf{X} - \boldsymbol{\mu})' \mathbf{U} \boldsymbol{\Lambda}^{-1} \mathbf{U}' (\mathbf{X} - \boldsymbol{\mu}) \\
&= \mathbf{Y}' \boldsymbol{\Lambda}^{\frac{1}{2}} \boldsymbol{\Lambda}^{-1} \boldsymbol{\Lambda}^{\frac{1}{2}} \mathbf{Y} \\
&= \mathbf{Y}' \mathbf{Y} \\
&= \sum_{i=1}^p Y_i^2 \sim \chi_p^2
\end{aligned}$$

2.

$$c^2 = \chi_2^2(0.1) = 4.605$$

After determining

$$\sigma_{12} = \rho_{12}\sqrt{\sigma_{11}}\sqrt{\sigma_{22}} = \frac{1}{2}\sqrt{2}\sqrt{1} = 2^{-\frac{1}{2}},$$

we have

$$\Sigma = \begin{bmatrix} 2 & 2^{-\frac{1}{2}} \\ 2^{-\frac{1}{2}} & 1 \end{bmatrix}$$

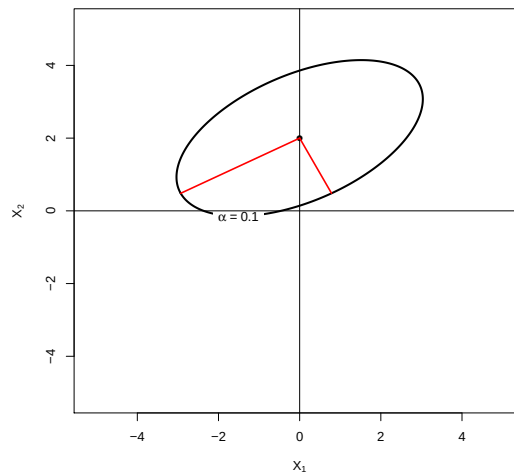
with eigenvalue-eigenvector pairs

$$\lambda_1 = 2.366 \quad \mathbf{e}_1 = [-0.888, -0.46]'$$

and

$$\lambda_2 = 0.634 \quad \mathbf{e}_2 = [0.46, -0.888]'$$

The ellipse will therefore be centred at $\boldsymbol{\mu} = (0, 2)$ with axes in the direction of $\mathbf{e}_1$ and $\mathbf{e}_2$, and with lengths (from origin) $c\sqrt{\lambda_1} = \sqrt{4.605}\sqrt{2.366} = 3.3$ and $c\sqrt{\lambda_2} = \sqrt{4.605}\sqrt{0.634} = 1.709$.



Homework exercise 3.3

$$\text{Let } Y = 3X_1 - 2X_2 + X_3 = \begin{bmatrix} 3 & -2 & 1 \end{bmatrix} \begin{bmatrix} X_1 \\ X_2 \\ X_3 \end{bmatrix} = \mathbf{a}'\mathbf{X}$$

$$\mathbf{a}'\boldsymbol{\mu} = \begin{bmatrix} 3 & -2 & 1 \end{bmatrix} \begin{bmatrix} 2 \\ -3 \\ 1 \end{bmatrix} = 13$$

$$\mathbf{a}'\Sigma\mathbf{a} = \begin{bmatrix} 3 & -2 & 1 \end{bmatrix} \begin{bmatrix} 1 & 1 & 1 \\ 1 & 3 & 2 \\ 1 & 2 & 2 \end{bmatrix} \begin{bmatrix} 3 \\ -2 \\ 1 \end{bmatrix} = 9$$

$$\therefore 3X_1 - 2X_2 + X_3 \sim N(13, 9)$$

Homework exercise 3.4

$$\Sigma = \begin{bmatrix} 1 & -2 & 0 \\ -2 & 5 & 0 \\ 0 & 0 & 2 \end{bmatrix}$$

1. X_1 and X_2 are not independent, since $\sigma_{12} = -2 \neq 0$.
2. X_2 and X_3 are independent, since $\sigma_{23} = 0$.
3. (X_1, X_2) and X_3 are independent, since $\Sigma_{12} = \mathbf{0}$, where Σ_{12} indicates the covariances between the partitions (X_1, X_2) and X_3 .
4. Let $Y = \begin{bmatrix} X_1 + X_2 \\ 2 \\ X_3 \end{bmatrix} = \mathbf{A}\mathbf{X}$ where $\mathbf{A} = \begin{bmatrix} \frac{1}{2} & \frac{1}{2} & 0 \\ 0 & 0 & 1 \end{bmatrix}$

$$\mathbf{A}\Sigma\mathbf{A}' = \begin{bmatrix} \frac{1}{2} & \frac{1}{2} & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & -2 & 0 \\ -2 & 5 & 0 \\ 0 & 0 & 2 \end{bmatrix} \begin{bmatrix} \frac{1}{2} & 0 \\ \frac{1}{2} & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} \frac{1}{2} & 0 \\ 0 & 2 \end{bmatrix}$$

Since the non-diagonals are zero, $\frac{X_1 + X_2}{2}$ and X_3 are independent.

Homework exercise 3.5

$$\mu_1 = 0, \mu_2 = 2, \sigma_{11} = 2, \sigma_{22} = 1, \rho_{12} = 0.5$$

$$\sigma_{12} = \rho_{12}\sqrt{\sigma_{11}\sigma_{22}} = 0.5\sqrt{(2)(1)} = \frac{1}{\sqrt{2}}$$

$$\begin{aligned} X_1|X_2 &\sim N\left(\mu_1 + \frac{\sigma_{12}}{\sigma_{22}}(x_2 - \mu_2), \sigma_{11} - \frac{\sigma_{12}^2}{\sigma_{22}}\right) \\ &\sim N\left(0 + \frac{1}{\sqrt{2}}(x_2 - 2), 2 - \frac{1}{2}\right) \\ &\sim N\left(\frac{1}{\sqrt{2}}(x_2 - 2), \frac{3}{2}\right) \end{aligned}$$

Homework exercise 3.6

1.

$$\mathbf{V}_i \sim N_p\left(\sum_{j=1}^4 c_j \boldsymbol{\mu}_j, \left(\sum_{j=1}^n c_j^2\right) \Sigma\right)$$

$$\text{For } \mathbf{V}_1 = \frac{1}{4}\mathbf{X}_1 - \frac{1}{4}\mathbf{X}_2 + \frac{1}{4}\mathbf{X}_3 - \frac{1}{4}\mathbf{X}_4, \mathbf{c}' = \begin{bmatrix} \frac{1}{4} & -\frac{1}{4} & \frac{1}{4} & -\frac{1}{4} \end{bmatrix}$$

such that

$$\begin{aligned} \sum_{j=1}^4 c_j \boldsymbol{\mu}_j &= \frac{1}{4}\boldsymbol{\mu}_1 - \frac{1}{4}\boldsymbol{\mu}_2 + \frac{1}{4}\boldsymbol{\mu}_3 - \frac{1}{4}\boldsymbol{\mu}_4 \\ &= \frac{1}{4}\boldsymbol{\mu} - \frac{1}{4}\boldsymbol{\mu} + \frac{1}{4}\boldsymbol{\mu} - \frac{1}{4}\boldsymbol{\mu} \\ &= \mathbf{0} \end{aligned}$$

and

$$\left(\sum_{j=1}^n c_j^2 \right) \Sigma = \frac{1}{4} \Sigma$$

$$\therefore \mathbf{V}_1 \sim N_p \left(\mathbf{0}, \frac{1}{4} \Sigma \right)$$

For $\mathbf{V}_2 = \frac{1}{4} \mathbf{X}_1 + \frac{1}{4} \mathbf{X}_2 - \frac{1}{4} \mathbf{X}_3 - \frac{1}{4} \mathbf{X}_4$, $\mathbf{c}' = \begin{bmatrix} \frac{1}{4} & \frac{1}{4} & -\frac{1}{4} & -\frac{1}{4} \end{bmatrix}$
such that

$$\begin{aligned} \sum_{j=1}^4 c_j \boldsymbol{\mu}_j &= \frac{1}{4} \boldsymbol{\mu}_1 + \frac{1}{4} \boldsymbol{\mu}_2 - \frac{1}{4} \boldsymbol{\mu}_3 - \frac{1}{4} \boldsymbol{\mu}_4 \\ &= \frac{1}{4} \boldsymbol{\mu} + \frac{1}{4} \boldsymbol{\mu} - \frac{1}{4} \boldsymbol{\mu} - \frac{1}{4} \boldsymbol{\mu} \\ &= \mathbf{0} \end{aligned}$$

and

$$\left(\sum_{j=1}^n c_j^2 \right) \Sigma = \frac{1}{4} \Sigma$$

$$\therefore \mathbf{V}_2 \sim N_p \left(\mathbf{0}, \frac{1}{4} \Sigma \right)$$

2.

$\mathbf{V}_1$ and $\mathbf{V}_2$ are jointly multivariate normal with covariance matrix

$$\begin{bmatrix} \left(\sum_{j=1}^n c_j^2 \right) \Sigma & (\mathbf{b}' \mathbf{c}) \Sigma \\ (\mathbf{b}' \mathbf{c}) \Sigma & \left(\sum_{j=1}^n b_j^2 \right) \Sigma \end{bmatrix}$$

Now $\mathbf{b}' \mathbf{c} = \sum_{j=1}^4 = 0 \implies \mathbf{V}_1$ and $\mathbf{V}_2$ are independent, and their joint density can be expressed as the product of their marginals:

$$\begin{aligned} f(\mathbf{v}_1, \mathbf{v}_2) &= f(\mathbf{v}_1) f(\mathbf{v}_2) \\ &= \frac{1}{(2\pi)^{\frac{p}{2}} \left| \frac{1}{4} \Sigma \right|^{\frac{1}{2}}} \exp \left[-\frac{1}{2} \mathbf{v}_1' 4 \Sigma^{-1} \mathbf{v}_1 \right] \times \frac{1}{(2\pi)^{\frac{p}{2}} \left| \frac{1}{4} \Sigma \right|^{\frac{1}{2}}} \exp \left[-\frac{1}{2} \mathbf{v}_2' 4 \Sigma^{-1} \mathbf{v}_2 \right] \\ &= \frac{1}{(2\pi)^p \left| \frac{1}{4} \Sigma \right|} \exp \left[-2 (\mathbf{v}_1' \Sigma^{-1} \mathbf{v}_1 + \mathbf{v}_2' \Sigma^{-1} \mathbf{v}_2) \right] \end{aligned}$$

3.

Let $\mathbf{V}_1 = \frac{1}{5} \mathbf{X}_1 + \frac{1}{5} \mathbf{X}_2 + \frac{1}{5} \mathbf{X}_3 + \frac{1}{5} \mathbf{X}_4 + \frac{1}{5} \mathbf{X}_5$ and $\mathbf{V}_2 = \mathbf{X}_1 - \mathbf{X}_2 + \mathbf{X}_3 - \mathbf{X}_4 + \mathbf{X}_5$
such that $\mathbf{c}' = \begin{bmatrix} \frac{1}{5} & \frac{1}{5} & \frac{1}{5} & \frac{1}{5} & \frac{1}{5} \end{bmatrix}$ and $\mathbf{b}' = \begin{bmatrix} 1 & -1 & 1 & -1 & 1 \end{bmatrix}$

$$\mathbf{E}(\mathbf{V}_1) = \sum_{j=1}^5 \frac{1}{5} \boldsymbol{\mu} = \boldsymbol{\mu}$$

$$\text{Cov}(\mathbf{V}_1) = \sum_{j=1}^5 \left(\frac{1}{5}\right)^2 \boldsymbol{\Sigma} = \frac{5}{25} \boldsymbol{\Sigma} = \frac{1}{5} \boldsymbol{\Sigma}$$

$$\mathbf{E}(\mathbf{V}_2) = \sum_{j=1}^5 c_j \boldsymbol{\mu} = \boldsymbol{\mu}$$

$$\text{Cov}(\mathbf{V}_2) = \left(\sum_{j=1}^5 c_j^2 \right) \boldsymbol{\Sigma} = 5 \boldsymbol{\Sigma}$$

$$\text{Cov}(\mathbf{V}_1, \mathbf{V}_2) = (\mathbf{b}' \mathbf{c}) \boldsymbol{\Sigma} = \frac{1}{5} \boldsymbol{\Sigma}$$
